Chapter 03
Numerical Initial Value Problem

Lesson 1

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3.1. Initial Value Problems

- Many engineering applications describe the evolution of some process with time.

In engineering and science, we often want to study **how something changes over time**.

For example:

- The temperature of a room changes with time.
- The speed of a moving vehicle changes with time.
- The amount of water in a tank changes with time.
- The population of bacteria changes with time.
- The voltage or current in an electrical circuit changes with time.

Mathematically, these changes are often described using a **differential equation**.

For example,

$$\frac{dy}{dt} = f(t, y)$$

means that the **rate of change of** $y$depends on the current time $t$and the current value of y.

Simple example

Suppose the velocity of a car changes according to

$$\frac{dv}{dt} = 2t.$$

This equation tells us **how the velocity changes with time**, but it does not tell us the actual velocity at a particular time.

- In order to define such an application, we require two distinct pieces of information: we need to know what the process is and also when or where the application started.
- To completely describe a problem, we generally need two types of information:

(a) What is the process?

This is described by the **differential equation**.

For example,

$$\frac{dy}{dt} = 2t$$

describes how $y$changes with time.

(b) When or where did the process start?

We also need to know the **initial condition**.

For example,

$$y(0) = 5.$$

This tells us that at time $t = 0$, the value of $y$was 5.

Therefore, we have:

$$\boxed{\frac{dy}{dt} = 2t, y(0) = 5}$$

The first equation describes the **process**, while the second tells us the **starting condition**.

Why is the initial condition important?

Consider

$$\frac{dy}{dt} = 2t.$$

If we integrate,

$$y = t^2 + C.$$

- There are infinitely many possible solutions because C can have different values.

But if we know

$$y(0) = 5,$$

then

$$5 = 0^2 + C$$

so

$$C = 5.$$

Therefore,

$$\boxed{y = t^2 + 5}$$

is the particular solution.

- So, **the differential equation tells us how the process behaves, while the initial condition tells us which particular solution represents the actual process.**

- In this section we begin with a discussion of some of these so-called initial value problem.

- Then we look at two numerical methods that can be used to approximate solutions of certain initial value problems.

What is an Initial Value Problem?

An **Initial Value Problem (IVP)** consists of:

1. A differential equation, and
2. An initial condition specified at a particular starting point.

The general form of a first-order IVP is

$$\boxed{\frac{dy}{dt} = f(t, y), y(t_0) = y_0}$$

where:

- t= independent variable, usually time
- y= dependent variable
- $f(t, y)$= function describing the rate of change
- t_0= initial time
- y_0= initial value

Example

$$\frac{dy}{dt} = 3y, y(0) = 2$$

Here,

$$\frac{dy}{dt} = 3y$$

is the **differential equation**, and

$$y(0) = 2$$

is the **initial condition**.

Together, they form an **initial value problem**

Numerical methods for Initial Value Problems

- Sometimes an IVP can be solved exactly using mathematical techniques.
- However, many real-world engineering problems involve complicated differential equations for which an exact solution is difficult or impossible to obtain.
- In such cases, we use **numerical methods** to obtain an approximate solution.

- Instead of finding the exact value of $y$at every point, numerical methods calculate approximate values such as

$$y(t_0), y(t_1), y(t_2), \ldots$$

For example, starting with

$$y(0) = 2,$$

we may use a numerical method to estimate

$$y(0.1), y(0.2), y(0.3), \ldots$$

- Two important numerical methods for solving first-order IVPs are:

Example:

Imagine that a car is travelling along a road. We want to know where the car will be after 10 seconds.

First: We need a mathematical rule describing how the car is moving.

For example,

$$\frac{ds}{dt} = v(t)$$

where $s$is the position and $v$is the velocity.

- **Second:** We need to know where the car was at the beginning.

For example,

$$s(0) = 100 \text{ m}.$$

Therefore,

$$\boxed{\frac{ds}{dt} = v(t), s(0) = 100}$$

is an **Initial Value Problem**.

Numerical methods for Initial Value Problems

1. Explicit method
2. Implicit method

1. Explicit method

Explicit = calculate the next value directly.

- In an **explicit method**, the next value y_{n+1} can be calculated **directly from values that are already known**.

A simple example is **Euler's method**:

$$\boxed{y_{n+1} = y_n + hf(t_n, y_n)}$$

Here:

y_n → known current value

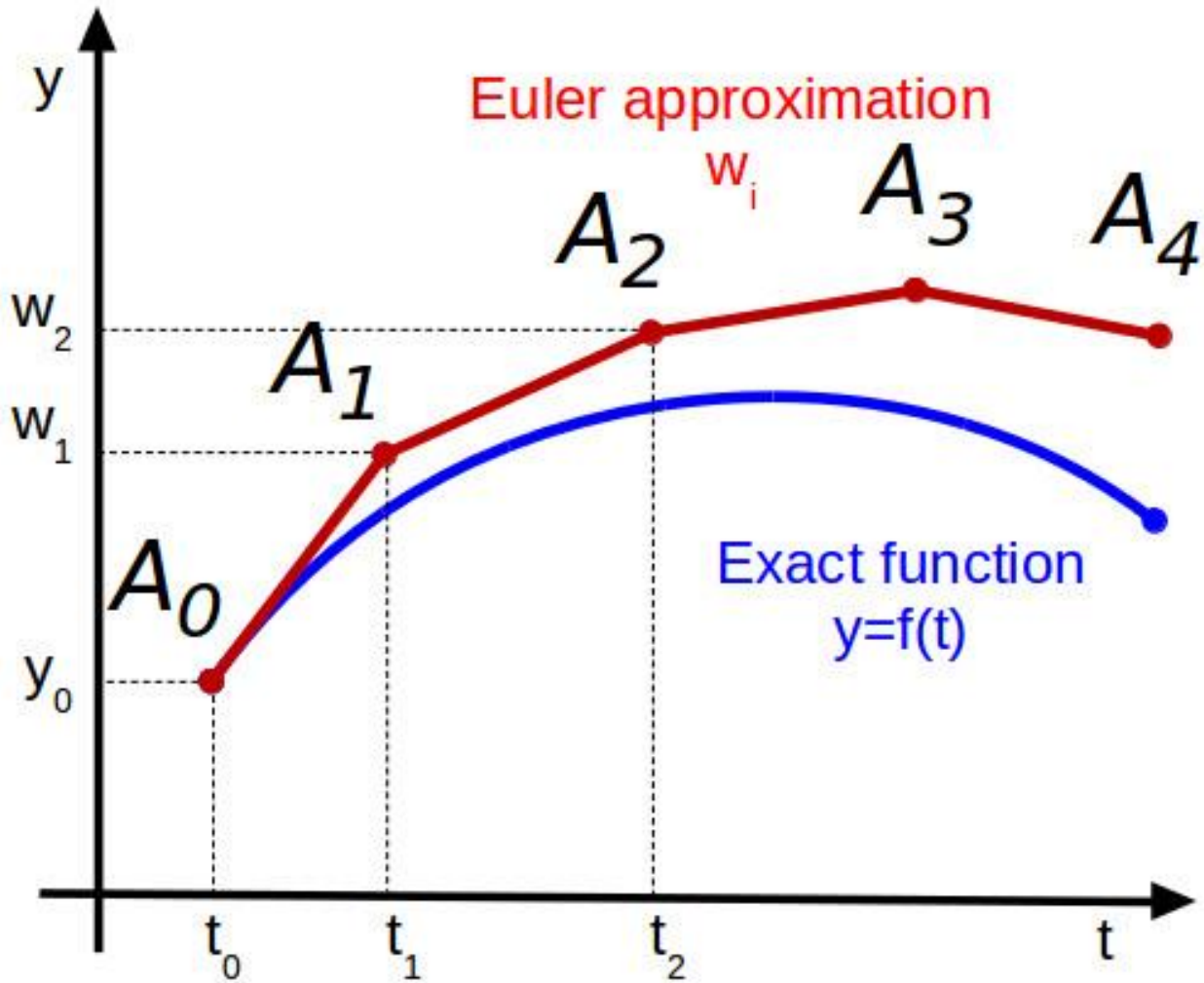
t_n → known current time

h → step size

$f(t_n, y_n)$ → calculated using known values

y_{n+1} → unknown value that we want to calculate

Notice that the right-hand side does **not contain** y_{n+1}.
Therefore, we can calculate it immediately.


y
Euler approximation
w_i
A_2
A_3
A_4
w_2
A_1
w_1
A_0
Exact function
y=f(t)
y_0
t_0
t_1
t_2
t

$t_n = t_0 + nh$ n is time step

$$y_{n+1} = y_n + hf(t_n, y_n)$$

$$y_1 = y_0 + hf(t_0, y_0)$$
$$y_2 = y_1 + hf(t_1, y_1)$$
$$y_3 = y_2 + hf(t_2, y_2)$$
$$y_4 = y_3 + hf(t_3, y_3)$$

$$t_1 = 0 + 1 \times h = h$$
$$t_2 = 0 + 2 \times h = 2h$$
$$t_3 = 0 + 3 \times h = 3h$$

Example

Consider

$$\frac{dy}{dt} = y, y(0) = 1$$

using Euler's method with

$$h = 0.1.$$

Euler's formula is

$$y_{n+1} = y_n + hf(t_n, y_n).$$

At the beginning:

$$t_0 = 0, y_0 = 1.$$

Therefore,

$$y_1 = 1 + (0.1)(1)$$

$$\boxed{y_1 = 1.1}$$

We obtained y_1 **directly**.

That's why Euler's method is an **explicit method**.

Question 1:

Suppose that $y = y(t)$is the solution to the initial value problem

$$\frac{dy}{dt} = -\frac{1}{(t+y)^2}, \qquad y(0) = 0.9$$

Carry out two-time steps of Euler's method with a step size of $h = 0.125$ so as to obtain approximations to $y(0.125)$ and $y(0.25)$.

Question 2:

Suppose that $y = y(t)$is the solution to the initial value problem

$$\frac{dy}{dt} = -y^2, \qquad y(0) = 0.5$$

Carry out two-time steps of Euler's method with a step size of $h = 0.01$ so as to obtain approximations to $y(0.01)$ and $y(0.02)$.

2. Implicit Method

- Another approach that can be used to address the initial value problem

$$\frac{dy}{dx} = f(t, y), \qquad y(0) = y_0$$

is to consider integrating the differential equation

$$\frac{dy}{dx} = f(t, y)$$

from $t = nh$ to $t = nh + h$. This leads to

$$[y(t)]_{t=nh}^{t=nh+h} = \int_{nh}^{(n+1)h} f(t, y)dt$$

that is

$$y(nh + h) - y(nh) = \int_{nh}^{(n+1)h} f(t, y)dt$$

And the problem now becomes one of approximating the integral on the right-hand side.

If we approximate the integral using the simple trapezium rule and replace the terms by their approximations we obtain the numerical method

$$y_{n+1} - y_n = \frac{1}{2}h(f_n + f_{n+1})$$

The procedure for time stepping with this method is much the same as that used for Euler's method. But with one difference. Let us imagine applying the method, we are given y_0 as the initial condition and now aim to find y_1 from

$$y_1 = y_0 + \frac{1}{2}h(f_0 + f_1)$$

$$= y_0 + \frac{1}{2}h\{f(0, y_0) + f(h, y_1)\}$$

And here is the problem: the unknown y_1 appears on both sides of the equation. We cannot, in general, find an explicit expression for y_1 and for this reason the numerical method is called an **implicit method**.

$$y_1 = y_0 + \frac{1}{2}h\{f(0, y_0) + f(h, y_1)\}$$

$$y_2 = y_1 + \frac{1}{2}h\{f(h, y_1) + f(2h, y_2)\}$$

$$y_3 = y_2 + \frac{1}{2}h\{f(2h, y_2) + f(3h, y_3)\}$$

$$y_4 = y_3 + \frac{1}{2}h\{f(3h, y_3) + f(4h, y_4)\}$$

Question 1

Suppose that $y = y(t)$is the solution to the initial value problem

$$\frac{dy}{dt} = \frac{1}{(t+1)}, \qquad y(0) = 1$$

Carry out two-time steps of trapezium method with a step size of $h = 0.2$ so as to obtain approximations to $y(0.2)$ and $y(0.4)$.

Question 2

Suppose that $y = y(t)$is the solution to the initial value problem

$$\frac{dy}{dt} = \frac{1}{(t^2 + 1)} - 2y, \qquad y(0) = 2$$

Carry out two-time steps of trapezium method with a step size of $h = 0.1$ so as to obtain approximations to $y(0.1)$ and $y(0.2)$.

Imagine you are climbing stairs.

Explicit

You know exactly where your current foot is and use that information to decide where to put your next foot.

$$\boxed{\text{Current information} \rightarrow \text{Next value}}$$

Implicit

Your next position is involved in the rule itself, so you have to **solve for where your next foot should be.**

$$\boxed{\text{Current information} + \text{Next value} \rightarrow \text{Solve for next value}}$$